

The Algorithmic Accountability Investigation: A Fairness Audit of COMPAS Recidivism Prediction

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Phase 1: Model and EDA

1.1 Domain Selection and Rationale

My Domain selection was intentional. I chose the criminal justice domain and the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) recidivism dataset because it represents one of the most consequential junctions of algorithmic decision-making and systemic inequality. Predictive risk assessment tools like COMPAS are embedded directly into sentencing and bail hearings across the United States, meaning their outputs carry legal weight over human lives. Unlike algorithmic bias in consumer contexts where a flawed recommendation might surface the wrong product, bias in criminal justice extends sentences, denies pretrial release, and determines whether someone goes home to their family or to a prison cell. The stakes are clear; they are measured in years of human freedom (Angwin et al., 2016).

In one documented case, an 18-year-old Black woman who stole a child's bicycle received a high-risk score and did not reoffend, while a white man with prior armed robbery convictions received a low-risk score and later committed additional crimes, expressing exactly how the algorithms error fall along racial lines (Angwin et al., 2016). This domain also is essential because it exposes a fundamental tension in algorithmic governance: the systems most urgently in need of independent auditing are often the ones operating with the least transparency. COMPAS's methodology was proprietary for years, shielded from public scrutiny while actively influencing judicial decisions. As algorithmic auditing becomes a formalized professional role under emerging regulations like the EU AI Act, criminal justice tools represent exactly the high-risk category that demands meticulous examination.

1.2 Who is Affected?

The defendants subject to COMPAS assessments are overwhelmingly low-income individuals who lack resources to contest algorithmic determinations (Angwin et al., 2016). Many are people of color navigating a system already shaped by racial disparities in policing and prosecution. A

high-risk label from COMPAS can mean denied bail, time in pretrial detention that costs people their jobs, housing, and custody of their children, before any conviction has occurred(Stevenson, 2018). The harm is immediate, material and disproportionately concentrated among communities that have historically had the least power.

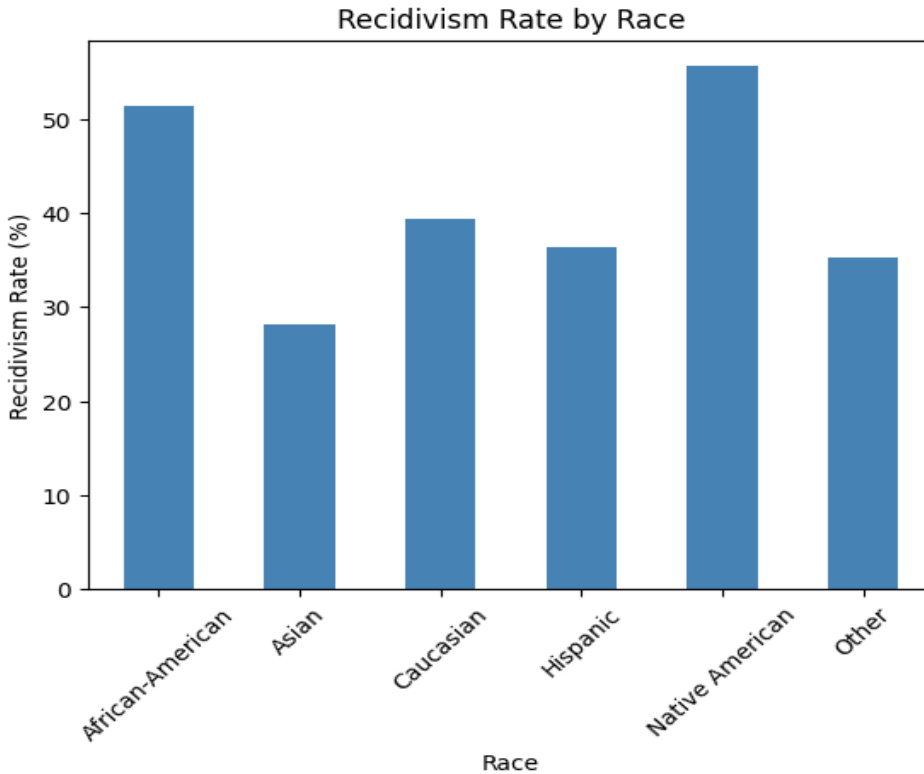
1.3 Exploratory Data Analysis (EDA)

The dataset contains 7,214 defendants screened in Broward County, Florida between 2013 and 2014. African- American defendants represent 51.2% of the sample, Caucasian defendants 34.0%, Hispanic 8.8%, and all other groups under 6% combined. Of all defendants, 45.1% reoffended within two years. COMPAS assigned Low risk scores to 54.0% of defendants, Medium to 26.6%, and High to 19.4%.

Table 1: Two Year Recidivism Rates by Race, Broward County (2013-2014)

Race	Recidivism Rate(%)
African- American	51.4
Asian	28.1
Caucasian	39.4
Hispanic	36.4
Native American	55.6
Other	35.3

Figure 1: Recidivism Rate by Race bar chart



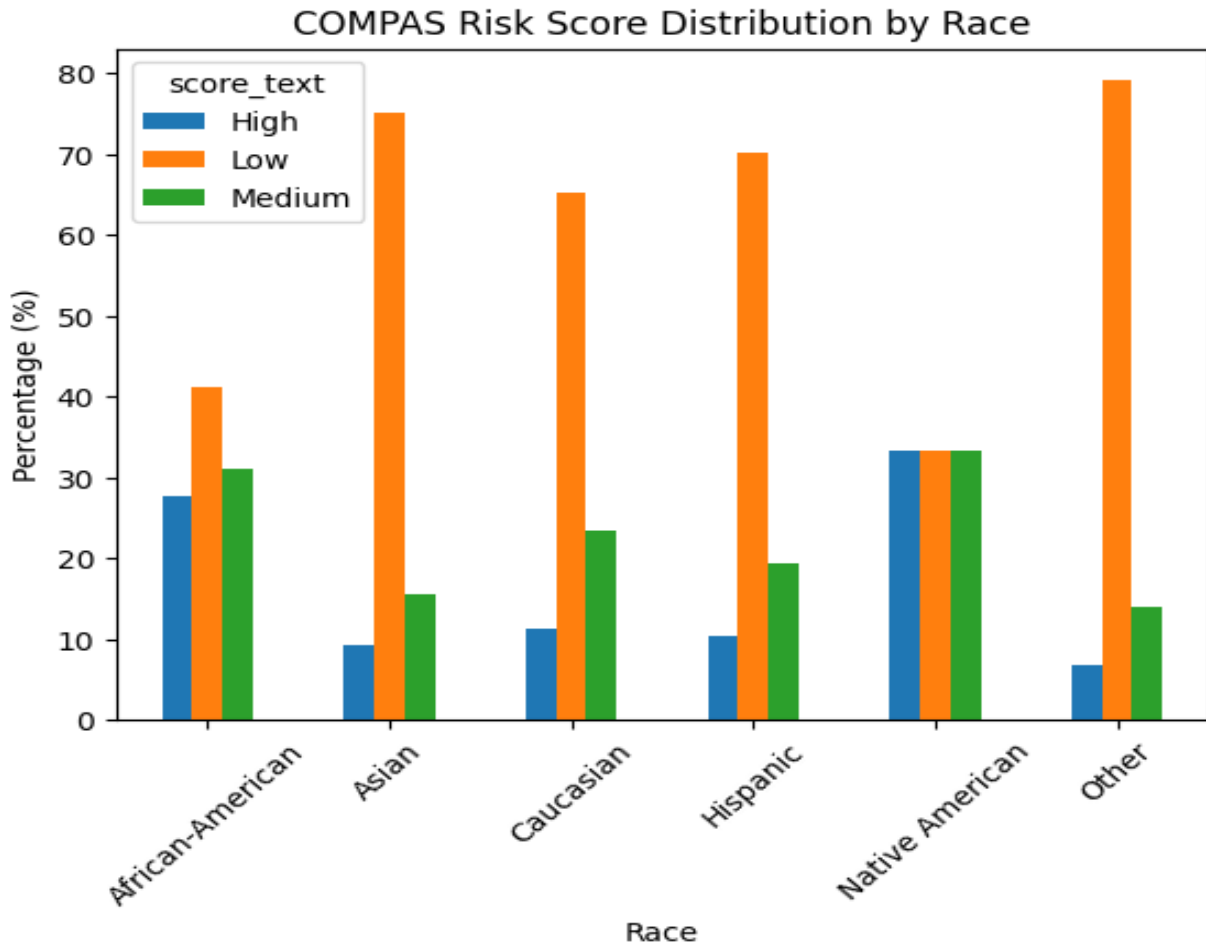
Two year Recidivism Rate by Race shows recidivism rates across all racial groups. African-American defendants reoffend at a rate of 51.4%, the highest among large groups, compared to 39.4% for Caucasian defendants. Native American defendants showed the highest rate at 55.6% but represents only 18 individuals in the dataset, making this statistically unreliable.

Table 2: COMPAS Risk Score Distribution by Race(%)

Race	High (%)	Medium(%)	Low(%)
African- American	27.7	31.1	41.2
Asian	9.4	15.6	75.0
Caucasian	11.2	23.5	65.2
Hispanic	10.5	19.3	70.2
Native American	33.3	33.3	33.3
Other	6.9	14.0	79.0

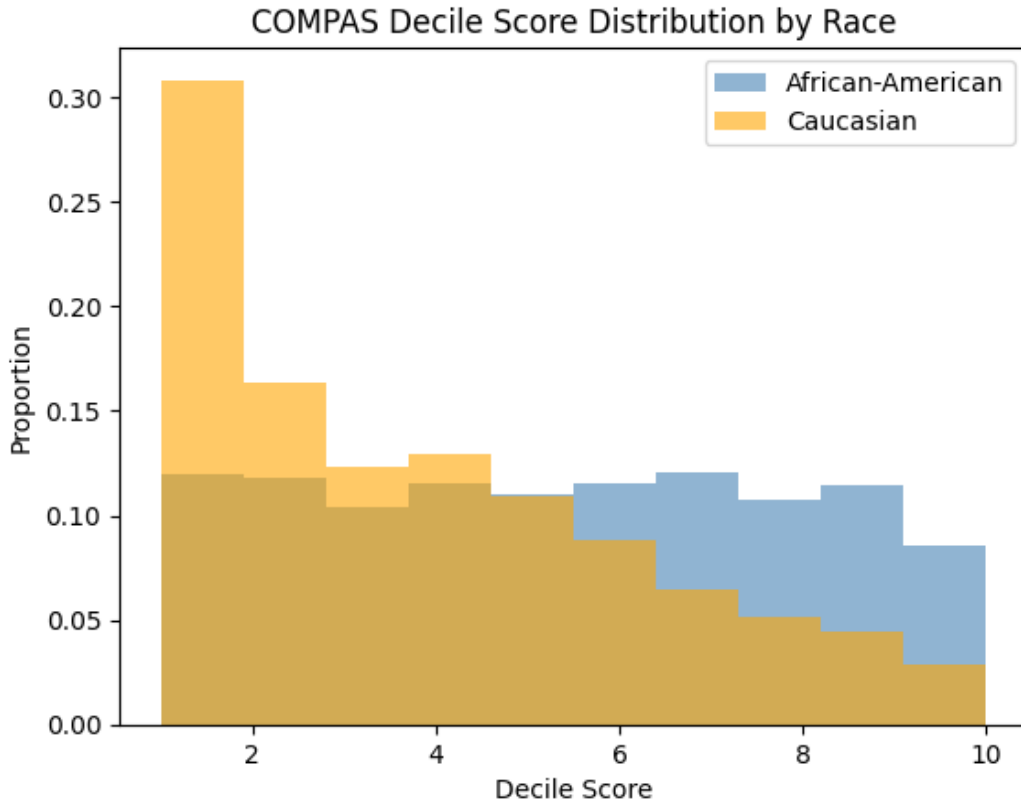
COMPAS risk label distribution by race. Native American n=18, statistically unreliable.

Figure 2: COMPAS Risk Score by Race



COMPAS Risk Score Distribution by Race. African-American defendants received High labels at 27.7%, compared to 11.2% for Caucasian defendants. Caucasian defendants received Low risk labels at 65.2% compared to only 41.2% for African-American defendants. This means Black defendants were labeled High risk at a more than double the rate of white defendants, while white defendants were cleared as Low risk at nearly 1.6 times the rate of Black defendants.

Figure 3: COMPAS Decile Score Distribution by Race



Caucasian defendants cluster heavily toward scores of 1 and 2, while African-American defendants show a flatter, more spread distribution extending into higher scores. This confirms that Black defendants are systematically pushed toward higher risk categories.

Table 3: Metrics Comparison (Recidivism Rates , Average COMPAS Scores, and High Risk Labels by Race).

Race	Recidivism Rate (%)	Avg. COMPAS Decile Score(1-10)	High Risk Label (%)
African-American	51.43	5.37	27.73
Asian	28.12	2.94	9.38
Caucasian	39.36	3.74	11.25
Hispanic	36.42	3.46	10.52
Native American	55.56	6.17	33.33
Other	35.28	2.95	6.9

Highlights African-American defendants received an average COMPAS decile score of 5.37 compared to 3.74 for Caucasian defendants, and were labeled High risk at 27.7% compared to 11.2% (Angwin et al., 2016)

1.4 Bias Prediction

Before training my own model, I recorded the following prediction: Based on the exploratory data analysis, I predict that the model will assign higher risk scores to African-American defendants even when controlling for factors like prior convictions and age. I believe this because race is included as a feature and the training data reflects pre-existing systemic disparities, so the model is likely to learn and reproduce from those patterns. I also expect the model to produce higher false positive rates for Black defendants, marking individuals as high risk who do not actually reoffend, reflecting the disparity already visible in COMPAS's own scores.

1.5 Model Training

A Random Forest classifier was trained using age, prior convictions, juvenile felony count, juvenile misdemeanor count, sex, and race as features. The dataset was split 80/20 into training and test sets using `random_state = 42` for reproducibility. Race and sex were one-hot encoded using `pd.get_dummies`. The training set contained 5,771 observations and the test set 1,443. Random Forest was selected because it handles mixed data types well, requires minimal preprocessing, and produces SHAP-compatible feature importance values needed for Phase 2 explainability analysis.

1.6 Baseline Classification Model Performance Metrics

Table 4: Baseline Classification Model Performance Metrics

	Precision	Recall	F1-Score	Support
0 (No reoffend)	0.69	0.70	0.69	823
1 (Reoffend)	0.59	0.58	0.59	620
Accuracy			0.65	1443
Macro Average	0.64	0.64	0.64	1443
Weighted Average	0.65	0.65	0.65	1443
AUC- ROC			0.68	

Random Forest classifier trained on COMPAS dataset (n=7,214). Test set n=1,443 (20% holdout, random_state=42). AUC-ROC of 0.68 indicates moderate discriminatory power. Class 0=did not reoffend within two years, Class 1=reoffended within two years.

The model achieved an overall accuracy of 65%, with an AUC-ROC of 0.68. Precision for non-reoffenders was 0.69 and recall 0.70. For reoffenders, precision was 0.59 and recall 0.58, indicating the model struggles more to correctly identify actual reoffenders than to correctly clear non-reoffenders. The lower recall for class 1 is significant in a criminal justice context, missing actual reoffenders means the model is simultaneously over-flagging low-risk Black defendants as shown in the fairness analysis in Phase 2.

Table 5: Disaggregated Model Accuracy by Race

Race	Accuracy	Count
African-American	0.647	731
Asian	0.600	5
Caucasian	0.651	505
Hispanic	0.632	117
Native American	0.667	3
Other	0.671	82

Accuracy computed on test set (n=1,443). Small sample sizes for Asian (n=5) and Native American (n=3) groups limit interpretability for those results.

Disaggregated by race, accuracy ranged from 60.0% for Asian defendants to 67.1% for Other. African-American defendants had an accuracy of 64.7%, slightly below the overall average. However, accuracy alone is misleading as a fairness measure. A model can be equally accurate across groups while still producing systematically different error types for different groups. Phase 2 will analyze this in depth through disaggregated false positive and false negative rates.

Phase 2: Fairness Audit and Explainability

2.1 Fairness Metrics Analysis

Table 6: Fairness Metrics Comparison: African-American vs. Caucasian

Metric	African-American	Caucasian
Demographic Parity	0.512	0.343
False Positive Rate	0.362	0.256
False Negative Rate	0.344	0.508
Predictive Parity	0.652	0.526

Fairness metrics computed on test set(n=1,443). Demographic parity=proportion predicted high risk. False positive rate=proportion of non-reoffenders wrongly labeled high risk, False negative rate=proportion of actual reoffenders labeled low risk. Predictive parity=accuracy when predicting high risk.

Three fairness metrics were computed and compared across African-American and Caucasian defendants. Demographic parity shows African-American defendants were predicted high risk at 51.2% compared to 34.3% for Caucasian defendants. The false positive rate for Black defendants was 36.2% compared to 25.6% for white defendants, meaning Black defendants who did not reoffend were wrongly labeled dangerous at a significantly higher rate. The false negative rate tells the opposite story: white defendants who did reoffend were let off as low risk at 50.8% compared to only 34.4% for Black defendants. Predictive parity shows that when the model does label someone high risk, it is actually more accurate for Black defendants at 65.2% compared to 52.6% for white defendants. Chouldechova(2017) analyzed the same Broward County dataset and found that without adjusting for any other variables, the odds of a non-reoffending Black defendant receiving a high-risk assessment were 2.6 times higher than for a white defendant in the same situation.

2.2 Metric Conflict and Ethical Framework

These three metrics cannot all be satisfied simultaneously. This is not a flaw in implementation. It is a mathematical impossibility formally demonstrated by Chouldechova (2017; Kearns & Roth, 2019). When base recidivism rates differ between racial groups, as they do in this dataset at 51.4% for Black defendants compared to 39.4% for white defendants, no classifier can achieve equal false positive rates, equal false negative rates, and equal predictive parity at the same time. Something has to give. The question then becomes not technical but rather ethical: which metric should we prioritize, and who bears the cost of that choice? Even after controlling for age, sex,

and numbers of priors, and charge severity, the odds of a non-reoffending Black defendant being wrongfully flagged as high risk remained 1.72 times higher than for a comparable white defendant (Chouldechova, 2017).

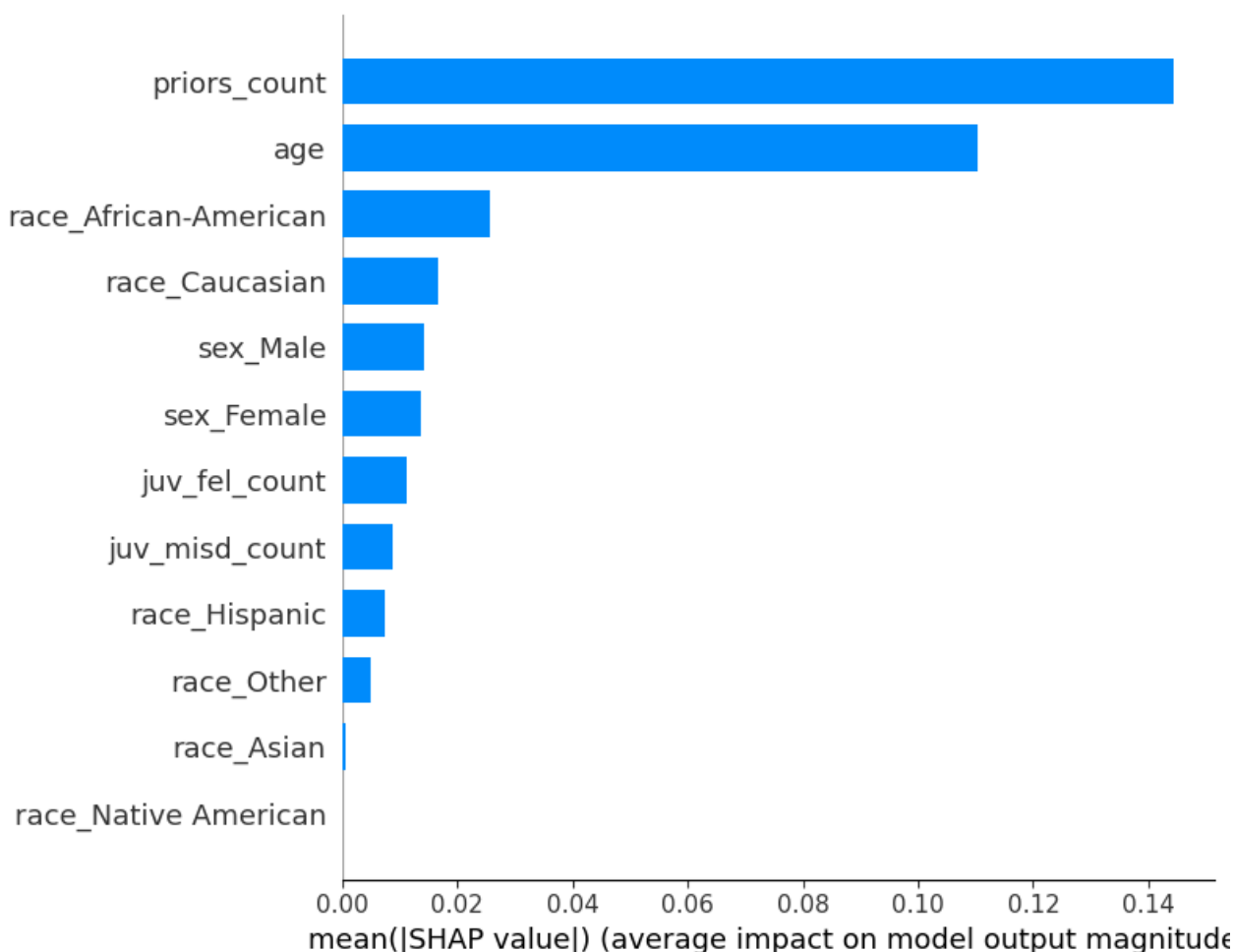
A utilitarian framework would prioritize overall predictive accuracy. The model performs at 65% accuracy overall with an AUC-ROC of 0.68, which maximizes correct predictions across the full population. Under this view, racial disparities are an acceptable tradeoff if the model produces the greatest predictive good for the greatest number. However, this framing obscures whose interests are being sacrificed. The greatest good calculus distributes the cost of false positives disproportionately onto Black defendants, who are already the most marginalized group in the criminal justice system.

A deontological framework reaches a different conclusion, Rooted in Kant's categorical imperative, deontological ethics holds that individuals must never be treated merely as means to an end, and that certain actions are intrinsically wrong regardless of outcomes. From this perspective, using race as a predictive feature is categorically impermissible regardless of any accuracy benefit. Every person has a right not to be judged by immutable characteristics they were born with. The SHAP analysis confirmed that `race_African-American` is the third most influential feature directly driving predictions. As a deontological auditor, it would be fair to conclude that no accuracy benefit justifies this.

I find the deontological framework more appropriate for this domain. Criminal justice is not a context where aggregate optimization is an acceptable standard. When the model's errors result in wrongful detention, extended sentences, or denied bail, the distribution of those errors matters as much as their frequency. A system that is 65% accurate overall but also systemically wrong about African-American defendants in the most harmful direction is not a fair system. It is an efficient one, and that is not the same.

2.3 Explainability Analysis

Figure 4: SHAP Feature Importance Bar Chart



Global SHAP Feature Importance. Mean absolute SHAP values computed on a random sample of 200 test observations(random_state=42). Higher values indicate greater average influence on model predictions.

Going into the SHAP analysis, I expected priors_count to dominate the model's predictions, but I did not anticipate how directly race would appear as an independent driver rather than hiding behind other variables. What the analysis revealed was more explicit than I predicted.

The SHAP analysis reveals that priors_count is by far the most influential feature driving high-risk predictions, followed by age as the second most influential, and race_African-American appearing explicitly as the third. This raises a deeper question that any serious audit must confront: what structural forces are acting as a benefactor behind these outcomes, quietly molding the model's behavior before a single prediction is made?

Starting with `priors_count`: while it carries the strongest predictive signal, it cannot be treated as a neutral variable. Prior convictions are not simply a measure of criminal behavior; they are also a measure of exposure to policing. Black communities in the United States have been historically and systematically over-policed, meaning Black defendants accumulate prior charges at higher rates not necessarily because they offend more, but because they are surveilled more. The data confirms this directly.

Table 7: Average Prior Conviction Count by Race

Race	Prior Convictions (mean)
Native American	6.00
African-American	4.44
Caucasian	2.59
Hispanic	2.25
Other	1.87
Asian	1.44

Mean prior conviction count by racial group. African-American defendants averaged 4.44 prior convictions compared to 2.59 for Caucasian defendants, a disparity of 71%.

African-American defendants averaged 4.44 prior convictions compared to 2.59 for Caucasian defendants, a gap of 71%. When the model learns from `priors_count`, it is learning from a variable that already encodes racial inequality. The bias does not begin inside the algorithm. It enters through the data long before training begins.

Table 8: Two-Year Recidivism Rate by Age Group

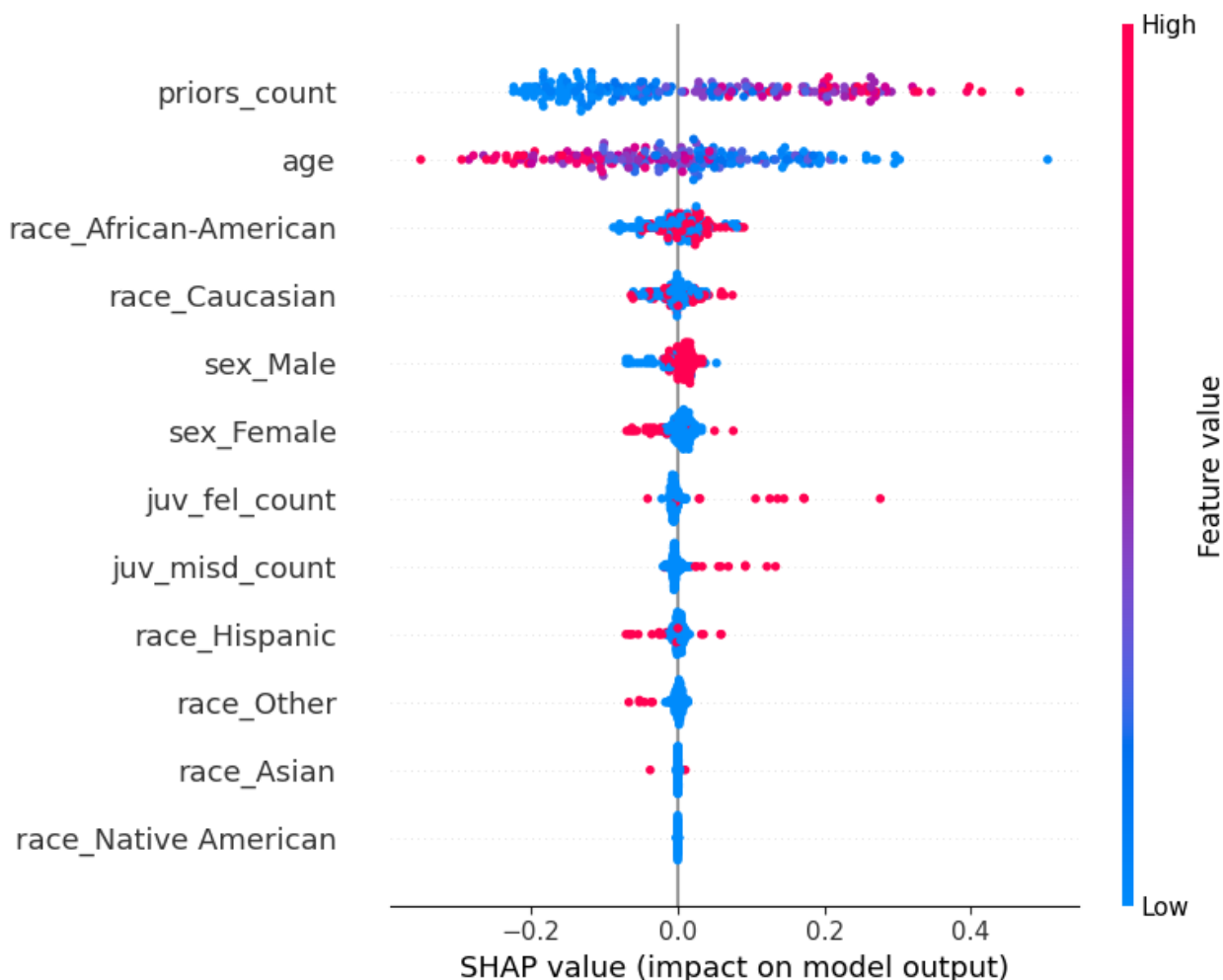
Age Category	Recidivism Rate (%)
Less than 25	56.5
25-45	45.9
Greater than 45	31.6

Recidivism rates by age group across full dataset (n=7,214). Younger defendants reoffend at significantly higher rates.

Age was the second most influential feature, and the data supports this as carrying legitimate predictive signal. Defendants under 25 reoffended at a rate of 56.5%, compared to 45.9% for those aged 25-45, and only 31.6% for those over 45. Younger defendants do reoffend at higher rates, which may reflect socioeconomic instability or lack of established social structures that

tend to stabilize with age. However, this cannot be read in isolation from race. The dataset skews younger for Black defendants, meaning age and race are correlated variables that compound each other's influence. A young Black defendant does not carry an age penalty and a race penalty separately. They interact and amplify each other within the model's decision structure.

Figure 5: SHAP Beeswarm Plot- Feature Direction and Magnitude



SHAP Beeswarm Plot showing directional influence of each feature on model predictions. Pink dots indicate high feature values, blue indicates low. Points to the right of center increase predicted risk; points to the left decrease it.

The beeswarm SHAP plot confirms the directional influence of race. Being identified as African-American directly pushes the model's prediction toward high risk, while being identified as Caucasian directly pushes it toward low risk. This is not an indirect effect operating through correlated variables. Race itself is acting as an independent lever pulling predictions in opposite

directions for Black and white defendants. The model learned this from training data that reflects decades of racially disparate criminal justice outcomes. In this sense, the algorithm is not malfunctioning, it is basically doing *exactly* what it was trained to do, and that is precisely the problem.

Comparing COMPAS's average risk scores against actual two-year recidivism rates further exposes the core of the system's bias.

Table 9: Average COMPAS Score vs. Actual Recidivism Rate by Race

Race	Average COMPAS Score (1-10)	Actual Recidivism(%)
African-American	5.37	51.4
Asian	2.94	28.1
Caucasian	3.74	39.4
Hispanic	3.46	36.4
Native American	6.17	55.6
Other	2.95	35.3

Average COMPAS decile scores compared to actual two-year recidivism rates by race. African-American defendants received scores 44% higher than Caucasian defendants despite a proportionally smaller gap in actual; reoffending rates.

African-American defendants received an average COMPAS decile score of 5.37 while their actual recidivism rate was 51.4%. Caucasian defendants received an average score of 3.74 while their actual recidivism rate was 39.4%. COMPAS assigned Black defendants scores disproportionately higher than their actual reoffending rates justify relative to white defendants

An intersectional analysis of false positives reveals that the burden of wrongful flagging falls intensely on African American Men.

Table 10: False Positives by Race and Gender

Race	Female	Male
African-American	7.0	123.0
Asian	0.0	1.0

Race	Female	Male
Caucasian	23.0	59.0
Hispanic	0.0	19.0
Native American	0.0	1.0
Other	1.0	12.0

False positives by race and gender in test set (n=1,443). False positive=defendant who did not reoffend but was predicted high risk.

Of all false positives in the test set, 123 were African-American males compared to 59 Caucasian males. Black women were also flagged as false positives as a disproportionate rate relative to their representation in the dataset, pointing to compounding disadvantages at the meeting point of race and gender.

2.4 Recourse Test

For this individual, a 22 year-old African-American male with 3 prior convictions who did not reoffend, no single change was sufficient to alter the model's high-risk prediction. Changing race alone, age alone, or prior convictions alone each left the prediction unchanged. Only by simultaneously setting prior convictions to zero and increasing age to 35 did the prediction change. Neither of these changes is actionable. A person cannot erase prior convictions or change their age. The only route to a favorable prediction requires changing characteristics that are either immutable or legally protected. Under GDPR Article 22, individuals subject to automated decisions have the right to clear explanation and a path to contest the outcome. The model provides neither. (GDPR Article 22, 2018)

2.5 Discovery Reflection

Before training the model, I predicted that race would influence predictions and that African-American defendants face higher false positive rates. I was directionally correct, but I underestimated how explicitly and independently race would operate as a feature. My assumption was that race would enter the model indirectly(absorbed through correlated variables like priors_count or age) and that removing or reducing those variables would largely neutralize the racial disparity. That assumption was wrong.

What the SHAP analysis revealed was that race_African-American is the third most influential feature operating directly and independently, not through proxies. I also assumed priors_count was a relatively clean predictor of risk. It was not, it is itself a racially skewed variable, reflecting differences in policing exposure rather than actual criminal behavior. The 71% gap in average prior convictions between Black and white defendants in this dataset was larger than I anticipated.

The recourse test produces the most surprising result. I assumed that reducing prior convictions alone would be enough to shift the model's prediction for a wrongly flagged individual. Even setting priors to zero had no effect in isolation. Only combining zero priors with a significantly older age changed the outcome, neither of which a 22-year-old can do. The model offered no realistic path to recourse, which I did not fully expect going in.

Phase 3: Regulatory and Policy Mapping

3.1 Regulatory Framework Overview

The audit findings from Phases 1 and 2 reveal a predictive model that directly uses race as a feature, produces racially disparate false positive rates, and offers no effective path to recourse

for affected individuals. These findings have significant policy implications. An expanding body of international, federal, and state-level law addresses exactly these harms. This section maps the audit’s technical findings to three specific regulatory frameworks: the EU AI Act(2024), the proposed U.S. Algorithmic Accountability Act (2023), and the Illinois Pretrial Fairness Act(2021, fully implemented 2023) and evaluates where the model falls short of legal compliance.

3.2 Regulation 1: EU AI Act (2024)– International

The European Union Artificial Intelligence Act, which entered into force in August 2024, represents the most comprehensive binding legal framework for AI systems in the world. It is directly and unambiguously applicable to recidivism prediction tools like COMPAS and the model audited here.

Annex III, Section 6(d)– High-Risk Classification

Annex III of the EU AI Act explicitly classifies as high-risk any AI system used by law enforcement authorities “for assessing the risk of a natural person offending or re-offending not solely on the basis of the profiling of natural persons.” This language directly encompasses recidivism prediction tools. The model audited here, which predicts two-year reoffending based on demographic and criminal history features, falls specifically within this classification. As a high-risk AI system, it would be subject to the full compliance obligations of Articles 9, 10, 13 and 14 before any deployment.

Article 9– Risk Management System

Article 9 requires that high-risk AI systems maintain a continuous risk management system throughout their lifecycle, including the identification and analysis of “known and reasonably foreseeable risks to health, safety or fundamental rights.” Article 9(8) specifically requires that testing be carried out “prior to being placed on the market or put into service” against “prior defined metrics and probabilistic thresholds.” The model audited here was never subjected to pre-deployment fairness testing. The false positive rate disparity, 36.2% for Black defendants versus 25.6% for white defendants (Table 6) represents exactly the type of foreseeable risk to

fundamental rights that Article 9 requires to be identified and mitigated before deployment. This model would be non-compliant with Article 9.

Article 10– Data and Data Governance

Article 10(2)(f) requires that training data be examined “in view of possible biases that are likely to... lead to discrimination prohibited under Union law,” and Article 10(2)(g) requires “appropriate measures to detect, prevent and mitigate possible biases identified.” The audit revealed two critical data governance failures. First, race was retained as a direct predictive feature despite its demonstrable influence as the third most important SHAP variable. Second, `priors_count`(the most influential variable) encodes structural over-policing disparities, with African-American defendants averaging 4.44 prior convictions compared to 2.59 for Caucasian defendants (Table 7). Article 10(2)(f) would require these biases to be identified before training. Article 10(2)(g) would require mitigation measures. Neither occurred. The model is non-compliant with Article 10.

Article 13– Transparency

Article 13(1) requires that high-risk AI systems be “sufficiently transparent to enable deployers to interpret a system’s output and use it appropriately,” Article 13(3)(b)(v) specifically requires documentation of the system’s performance “regarding specific persons or groups of persons on which the system is intended to be used.” COMPAS operated as a proprietary black box for years, providing no disaggregated performance metrics to judicial deployers. Judges applying COMPAS scores had no access to information showing that the system produced false positives at 41% higher rates for Black defendants (Table 6) than white defendants. Article 13(3)(b)(ii) further requires documentation of known accuracy limitations (information that was never provided). This model is non-compliant with Article 13.

Article 14– Human Oversight

Article 14(4)(b) requires that human overseers remain “aware of the possible tendency of automatically relying or over-relying on the output produced by a high-risk AI system,” specifically addressing automation bias in systems used to inform decisions by natural persons. Article 14(4)(d) requires that deployers be able “to decide, in any particular situation, not to use

the high-risk AI system or to otherwise disregard, override or reverse the output.” In practice, COMPAS scores have been treated as authoritative by courts with limited human review. No technical or procedural mechanism was built into the system to flag or counteract automation bias. The model is non-compliant with Article 14.

3.3 Regulation 2: Proposed Algorithmic Accountability Act (2023)– U.S. Federal Government

The Algorithmic Accountability Act of 2023 was introduced in the U.S. Congress to require companies and government entities deploying automated decision systems to assess their impacts on accuracy, fairness, bias, and discrimination. While not yet enacted into law, the Act establishes the clearest articulation of the direction of U.S. federal algorithmic governance and provides an important benchmark for evaluating this model’s compliance readiness.

The Act would specifically target “automated critical decision processes,” systems making decisions affecting health, finances, employment, housing, and criminal justice outcomes. Its core requirement is that companies assess the impacts of automating critical decision-making, “including decision processes that have already been automated.” This retroactive reach is important because it would apply to COMPAS even though it has been deployed for over a decade.

The Act explicitly identifies as problematic the “improper use of a person’s race... in making critical decisions.” The SHAP analysis confirmed that race_African-American is the third most influential predictive feature in this model, constituting exactly the improper racial influence the Act points to.

The Act would also require the FTC to establish a public repository showing “which critical decisions have been automated by companies along with information such as data sources, high level metrics and how to contest decisions.” The recourse test demonstrated that no contestation pathway exists for individuals wrongly labeled high risk directly conflicting with this transparency requirement.

Under this framework, the model would be non-compliant on three conditions: no impact assessment was conducted, race is used as a direct predictive feature, and no contestation mechanism exists for affected defendants.

3.4 Regulation 3: Illinois Pretrial Fairness Act(Public Act 101-652, 2023)– State Level

The Illinois Pretrial Fairness Act, signed into law in February 2021 and fully implemented in September 2023, abolishes cash bail and reformed pretrial detention in Illinois. This legislation is relevant to the COMPAS audit because COMPAS has been used in Illinois courts to inform pretrial release decisions(exactly the context the Act reformed).

The Act mandates that pretrial detention decisions be based on individualized assessments of risk rather than algorithmic scores treated as determinative. It requires courts to consider specific, documented factors related to the individual defendant rather than relying on aggregate risk prediction. This creates a direct tension with COMPAS-style tools that produce a single decile score without providing judges with interpretable, individualized explanations of what drove that score.

The audit’s recourse test finding is relevant here. A 22-year-old Black defendant with three prior convictions who posed no actual reoffending risk was labeled as high risk with no actionable path to change that outcome. Under the Illinois Pretrial Fairness Act, this type of algorithmic determination, without individualized explanation or judicial review is inconsistent with the Act’s mandate for individualized pretrial assessment. The Act supports the principle that risk tool outputs must be transparent and contestable, standards this model fails to meet.

3.5 Compliance Gap Analysis

#	Finding from Audit	Regulatory Requirement	Regulation and Section	Compliance Status	Explanation
1	Race is 3 rd most influential predictive feature	Examination and mitigation of bias in training data	EU AI Act Article 10(2)(f)(g)	Non-compliant	Race directly drives predictions; Article 10 requires bias identification and mitigation before training.

2	False positive rate 41% higher for Black defendants	Pre-deployment impact assessment required	Algorithmic Accountability Act (2023)	Non-compliant	Disparate FPR requires documented assessment; none was conducted
3	No pre-deployment bias audit conducted	Risk management system required before deployment	EU AI Act Article 9(8)	Non-compliant	No testing against fairness metrics conducted prior to deployment
4	No recourse path for wrongly flagged individuals	Right to explanation and contestation of automated decisions	GDPR Article 22(2018)	Non-compliant	Recourse test shows no actionable path to change prediction exists.
5	priors_count encodes racial over-policing disparities	Representative and unbiased training data required.	EU AI Act Article 10(2)(f)	Non-compliant	71% gap in prior convictions reflects surveillance inequality not actual behavior.
6	No disaggregated performance metrics provided to judges	Performance by demographic group must be disclosed	EU AI Act Article 13(3)(b)(v)	Non-compliant	Judges never informed that false positive rates differ by race
7	No automation bias protections for judicial users	Human oversight must address automation bias	EU AI Act Article 14(4)(b)	Non-compliant	No mechanism exists to prevent over-reliance on COMPAS scores
8	No individualized explanation provided to defendants	Individualized pretrial assessment requirement	Illinois Pretrial Fairness Act (2023)	Non-compliant	Model produces single decile score with no individualized explanation
9	Demographic parity gap of 16.9 percentage points	Impact assessment must document disparate outcomes	Algorithmic Accountability Act (2023)	Non-compliant	Prediction rate gap between racial groups requires documentation
10	Intersectional disparity: Black men flagged at 2x rate compared	Assessment must cover protected class intersections	Algorithmic Accountability Act (2023)	Non-compliant	Intersectional false positive analysis shows compounded racial and gender harm

	to white counterparts				
11	No contestation mechanism for affected defendants	Public access to contestation information required	Algorithmic Accountability Act (2023)	Non-compliant	Act requires repository showing how to contest automated decisions
12	Judicial ability to override system output unclear	Deployer must be able to disregard system output	EU AI Act Article 14(4)(d)	Gray area	Judges technically can override but no formal mechanism or training exists

3.6 Regulatory Comparison

To show how different jurisdictions treat the same finding, this section examines the false positive rate disparity. Black defendants who did not reoffend were wrongly labeled high risk at 36.2% compared to 25.6% for white defendants, examined through three distinct regulatory lenses.

GDPR and EU AI Act (European Union) The EU framework takes the most substantive and preventative approach. GDPR Article 22 gives affected individuals the right to human review and a clear explanation when automated decisions produce legal effects. The EU AI Act extends this matter by requiring under Articles 9 and 10 that the disparity be identified and mitigated before deployment. Crucially, Article 13(3)(b)(v) requires that performance across demographic groups be disclosed to judicial deployers, meaning judges would need to be informed that the system produces false positives at 41% higher rates for Black defendants before they use it. The EU framework is the only one of the three that would have prevented this model from reaching deployment in its current form.

United States– Proposed Algorithmic Accountability Act. The U.S. approach is process-oriented rather than outcome-oriented. The act would require a documented impact assessment showing the false positive rate disparity but does not specify what level of disparity is acceptable or mandate remediation. A deploying agency could theoretically document the 41% gap and proceed with deployment as long as the assessment was filed with the FTC. The Act's greatest

strength is transparency, requiring public disclosure of automated decisions and how to contest them. Its greatest limitation is the absence of binding numerical fairness standards or enforcement mechanisms requiring actual reduction in disparate harm.

Illinois– Pretrial Fairness Act. Illinois takes a structural rather than technical approach. Rather than setting fairness thresholds for algorithmic tools, the Act removes algorithmic scores from the center of pretrial decisions entirely, requiring individualized judicial assessment. This sidesteps the fairness metric conflict identified in Phase 2: if no single algorithmic score determines pretrial detention, the mathematical impossibility of simultaneously satisfying demographic and predictive parity becomes less legally consequential. The Illinois approach is arguably the most protective for defendants but leaves open questions about what replaces algorithmic assessment and whether unaided judicial discretion produces fairer outcomes in practice.

Where they agree: All three frameworks recognize that automated systems used in high-stakes criminal justice decisions require transparency, documentation, and accountability mechanisms. All three would require that defendants be informed when an automated system influenced a decision affecting their liberty.

Where they diverge: The EU framework mandates pre-deployment prevention with specific technical requirements. The U.S. Federal framework mandates post-deployment documentation without outcome standards. Illinois mandates structural reform rather than technical compliance. The EU approach is binding law. The U.S. Act remains proposed legislation. The Illinois Act is enacted but domain specific.

What is left unaddressed: None of the three frameworks provides a retroactive remedy for individuals already harmed by COMPAS's racial disparities before these regulations existed. Thousands of defendants received inflated risk scores over the past decade with no legal recourse. Additionally, none establishes binding numerical thresholds for acceptable false positive rate disparities across racial groups, leaving the 41% gap documented in this audit in a legal gray area where harm is acknowledged but no specific standard has been violated.

Phase 4: Responsible AI Governance Framework

4.1 Introduction

The findings of this audit uncover how a recidivism prediction model trained on the COMPAS dataset produces racially disparate outcomes, uses race as a direct predictive feature and provides no clear path to recourse for affected individuals. Every item in the pre-deployment checklist and every element of the stakeholder disclosure document is grounded in a specific finding from this audit or a specific regulatory requirement identified in Phase 3. Organizations deploying predictive models in criminal justice settings should treat this framework as a minimum standard, not a ceiling.

4.2 Pre-Deployment Checklist

Table 12: Pre-Deployment Checklist for Criminal Justice Predictive Models

#	Action Item	Specific Findings or Regulatory Basis	Verification Method
1	Remove race as a direct input feature and retest model performance	SHAP analysis confirmed race_African-American is the 3 rd most influential feature; EU AI Act Article 10(2)(f)	Rerun SHAP analysis and confirm race features no longer appear in top 10
2	Compute and document disaggregated fairness metrics across all racial groups before deployment	FPR gap of 36.2% vs 25.6% between Black and white defendants; Algorithmic Accountability Act (2023)	Produce Table 6- equivalent fairness metrics report signed by an independent reviewer
3	Conduct a recourse test on at least 20 false positive cases	Recourse test showed no single actionable change flipped prediction for a wrongly flagged individual; GDPR Article 22	Document which changes flip predictions and whether those changes are actionable
4	Examine priors_count and all criminal history variables for racial proxy effects	priors_count showed a 71% gap between Black and white defendants reflecting over policing; EU AI Act Article 10(2)(f)	Run correlation analysis between priors_count and race; document findings
5	Provide all judicial deployers with a written disaggregated performance summary before use	Judges had no access to information showing race-based FPR differences; EU AI Act Article 13(3)(b)(v)	Obtain signed acknowledgment from each judicial deployer confirming receipt
6	Establish an automation bias safeguard. Require judges to form an independent assessment before viewing any algorithmic score	EU AI Act Article 14(4)(b) requires awareness of automation bias risk	Track and audit divergence rates between judicial assessments and model scores quarterly

7	Commission an independent third-party bias audit including intersectional analysis	Intersectional analysis showed Black men were flagged as false positives at more than double the rate of white men; Algorithmic Accountability Act (2023)	Publish audit report publicly before deployment
8	Create a formal contestation mechanism for defendants	No contestation pathway existed for any defendant in this audit; GDPR Article 22; Illinois Pretrial Fairness Act	Document the contestation process, test it with at least 5 simulated cases, and make it accessible to defendants without legal representation
9	Document the fairness metric conflict and publicly justify which metric is prioritized	Demographic parity, FPR balance, and predictive parity cannot all be satisfied simultaneously when base rates differ; Chouldechova (2017); (Kearns & Roth, 2019)	Produce a written policy statement explaining which metric was prioritized and why, reviewed by an ethics board
10	Complete the NIST AI Risk Management Framework Govern, Map, Measure, and Manage functions	NIST AI RMF (2023) recommends these four functions as the foundation of responsible AI governance	Produce completed NIST AI RMF documentation before deployment sign-off
11	Establish post-deployment monitoring. Track FPR by race quarterly	EU AI Act Article 9 requires continuous iterative risk management throughout the system lifecycle	Trigger mandatory review if FPR gap between any two racial groups increases by more than 5 percentage points from baseline
12	Require plain-language individualized explanations for every high-risk prediction	Illinois Pretrial Fairness Act mandates individualized assessment; GDPR Article 22 requires clear explanation	Test explanation system with 10 non-technical participants and confirm they can understand and act on the explanation

Pre-deployment checklist for criminal justice predictive models. Each item references a specific audit finding or regulatory requirement. All items must be completed and documented before deployment.

4.3 Stakeholder Communication Template

The following document is designed to be understood by all persons subject to an algorithmic risk assessment, regardless of their educational background or familiarity with technology. No technical knowledge is assumed.

NOTICE TO DEFENDANT: A COMPUTER PROGRAM WAS USED IN YOUR CASE

What happened? A computer program called a risk assessment tool was used to generate a score that may have influenced decisions about your case, including decisions about bail or pretrial release. You have the right to know this occurred and to understand what it could mean for you.

What does this program do? The program looks at information about you. This includes your age, your criminal history, and other personal identifiable details and produces a number between 1 and 10. A higher number means the program thinks there is a greater chance you will be arrested for a new crime within the next two years. Or a lower number means a smaller chance. This number is referred to as a risk score.

What information was used about you? Your age, prior arrest and conviction history, juvenile record, and sex. Your race was included as a factor. Independent testing has shown this directly and unfairly inflates risk scores for African- American defendants.

Has this program been tested for fairness? An independent audit of this type of program found that it incorrectly labeled people who did not reoffend as high risk at a higher rate for Black defendants than for white defendants. Specifically, Black defendants who did not go on to commit a new crime were wrongly labeled high risk at a rate 41% higher than white defendants in the same situation. The program has not been found to be fully fair across racial groups.

Does this score decide my case? No. A judge makes the final decision about your case. The risk score is one piece of information the judge may consider, but it does not and should not be the only factor. You have the right to ask what weight, if any, was given to this score in your case.

What can you do if you believe this score is wrong? You have the right to contest your risk score. Here is how:

1. Ask for your score and the reasons behind it in writing.
2. Request a human review. You can ask that a human review your case individually before any decision is made based on the score.
3. Tell your attorney. You have the right to communicate that a risk score was used to challenge it in court.
4. If you do not have an attorney, contact the public defender's office in your county. You have the right to legal representation.

Questions? Contact the court clerk's office or your attorney. You have the right to ask questions and receive plain language responses.

Prepared in compliance with GDPR Article 22, the Illinois Pretrial Fairness Act(Public Act 101-652), and the proposed Algorithmic Accountability Act (2023).

References

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Code Appendix

Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn, shap

Data source: ProPublica COMPAS Analysis GitHub Repository. Data loaded directly via URL: <https://raw.githubusercontent.com/propublica/compas-analysis/master/compas-scores-two-years.csv>

Reproducibility note: All random states are set to `random_state = 42`. Running all cells from top to bottom will reproduce every number cited in this report.

IS467_Final_Capstone_angelag5.ipynb_

```
[108]
0s
import pandas as pdimport numpy as npimport matplotlibimport matplotlib.pyplot as
pltimport seaborn as snsimport sklearnfrom sklearn.ensemble import
RandomForestClassifierfrom sklearn.model_selection import train_test_splitfrom
sklearn.metrics import classification_report, roc_auc_scoreimport shap# Verify
versionsprint("Library Versions:")print(f"pandas: {pd.__version__}")print(f"numpy:
{np.__version__}")print(f"sklearn: {sklearn.__version__}")print(f"shap:
{shap.__version__}")print(f"matplotlib: {matplotlib.__version__}")print(f"seaborn:
{sns.__version__}")

Library Versions:
pandas: 2.2.2
numpy: 2.0.2
sklearn: 1.6.1
shap: 0.51.0
matplotlib: 3.10.0
seaborn: 0.13.2

[109]
0s
# loads data directly from ProPublica GitHub - no local file needed!df =
pd.read_csv(url)url = "https://raw.githubusercontent.com/propublica/compas-
analysis/master/compas-scores-two-years.csv"df

[110]
0s
#Distribution of key demographic and outcome variablesdf_counts = {    'race':
df['race'].value_counts(),    'sex': df['sex'].value_counts(),    'age_cat':
df['age_cat'].value_counts(),    'two_year_recid':
df['two_year_recid'].value_counts(),    'score_text': df['score_text'].value_counts()}

[111]
0s
df_counts
{'race': race
African-American    3696
Caucasian          2454
Hispanic            637
Other               377
Asian               32
Native American     18
Name: count, dtype: int64,
'sex': sex
Male                5819
Female              1395
Name: count, dtype: int64,
'age_cat': age_cat
```

```

25 - 45          4109
Greater than 45   1576
Less than 25     1529
Name: count, dtype: int64,
'two_year_recid': two_year_recid
0          3963
1          3251
Name: count, dtype: int64,
'score_text': score_text
Low         3897
Medium      1914
High        1403
Name: count, dtype: int64}

[112]
0s
# Calculates racial breakdown as percentages for EDA
df_race_pct = df['race'].value_counts(normalize=True) * 100
df_race_pct

[113]
0s
df_recid_pct = df['two_year_recid'].value_counts(normalize=True) * 100
df_recid_pct

[114]
0s
df_score_pct = df['score_text'].value_counts(normalize=True) * 100
df_score_pct

[115]
0s
black_def = df[df['race'] == 'African-American']
white_def = df[df['race'] == 'Caucasian']

[116]
0s
# Creates binary high risk indicator for groupby analysis
df['is_high_risk'] = (df['score_text'] == 'High')

df_comparison = df[df['race'].isin(['African-American',
'Caucasian', 'Asian', 'Hispanic', 'Native American', 'Other' ])].groupby('race').agg(
    recidivism_rate=('two_year_recid', 'mean'),
    avg_compas_score=('decile_score', 'mean'),
    high_risk_pct=('is_high_risk', 'mean')
)

# Compare recidivism rates, COMPAS scores, and high risk labels by race
df_comparison['recidivism_rate'] = (df_comparison['recidivism_rate'] * 100)
df_comparison['avg_compas_score'] = df_comparison['avg_compas_score']
df_comparison['high_risk_pct'] = (df_comparison['high_risk_pct'] * 100)

df_comparison.round(2)

[117]
0s
recid_by_race = df.groupby('race')['two_year_recid'].mean() * 100

# Visualization 1: Recidivism rate by race

```

```
recid_by_race.plot.bar(title='Recidivism Rate by Race', ylabel='Recidivism Rate (%)',
xlabel='Race', color='steelblue', rot=45)
```

```
[118]
0s
import matplotlib.pyplot as plt
# Visualization 3: COMPAS decile risk score distribution by race
# shows black defendants cluster toward higher scores than white defendants
fig, ax = plt.subplots()

df[df['race'] == 'African-American']['decile_score'].plot.hist(
    ax=ax, alpha=0.6, bins=10, label='African-American', color='steelblue',
    density=True
)
df[df['race'] == 'Caucasian']['decile_score'].plot.hist(
    ax=ax, alpha=0.6, bins=10, label='Caucasian', color='orange', density=True
)

ax.set_title('COMPAS Decile Score Distribution by Race')
ax.set_xlabel('Decile Score')
ax.set_ylabel('Proportion')
ax.legend()
plt.show()
```

```
[119]
0s
score_by_race =
df.groupby('race')['score_text'].value_counts(normalize=True).unstack() * 100
# Visualization 2: COMPAS risk score distribution by race
score_by_race.plot.bar(title='COMPAS Risk Score Distribution by Race',
ylabel='Percentage (%)', xlabel='Race', rot=45)
```

```
[120]
0s
score_by_race =
df.groupby('race')['score_text'].value_counts(normalize=True).unstack() * 100
score_by_race
```

Next steps:

```
[121]
0s
# Select features for Random Forest model
# race and sex are one-hot encoded via pd.get_dummies
features = ['age', 'priors_count', 'juv_fel_count', 'juv_misd_count', 'sex', 'race']
target = 'two_year_recid' #Binary: 0 = no reoffend, 1 = reoffend within 2 years

df_model = df[features + [target]].dropna()

df_model = pd.get_dummies(df_model, columns=['sex', 'race'])

print(df_model.shape)
print(df_model.columns.tolist())
(7214, 13)
['age', 'priors_count', 'juv_fel_count', 'juv_misd_count', 'two_year_recid',
'sex_Female', 'sex_Male', 'race_African-American', 'race_Asian', 'race_Caucasian',
'race_Hispanic', 'race_Native American', 'race_Other']
```

```
[122]
0s
```

```

from sklearn.model_selection import train_test_split

features = ['age', 'priors_count', 'juv_fel_count', 'juv_misd_count', 'sex', 'race']
target = 'two_year_recid'

df_model = df[features + [target]].dropna()
df_model = pd.get_dummies(df_model, columns=['sex', 'race'])

X = df_model.drop(columns=[target])
y = df_model[target]

#80/20 train/test split with fixed random state for reproducibility
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

print("Training size:", X_train.shape)
print("Test size:", X_test.shape)
Training size: (5771, 12)
Test size: (1443, 12)

[123]
0s
from sklearn.ensemble import RandomForestClassifier

# Train Random Forest - 100 trees, random_state=42
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
print("Model trained!")
Model trained!

[124]
0s
from sklearn.metrics import classification_report, roc_auc_score

y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

# Baseline performance metrics - overall and disaggregated by race
print(classification_report(y_test, y_pred))
print("AUC-ROC:", roc_auc_score(y_test, y_prob))

```

	precision	recall	f1-score	support
0	0.69	0.70	0.69	823
1	0.59	0.58	0.59	620
accuracy			0.65	1443
macro avg	0.64	0.64	0.64	1443
weighted avg	0.65	0.65	0.65	1443

```

AUC-ROC: 0.6801885313369654

[125]
0s
# Add predictions back to test set for group analysis
X_test_copy = X_test.copy()
X_test_copy['y_true'] = y_test.values
X_test_copy['y_pred'] = y_pred

# Accuracy per racial group
def accuracy_by_group(df, race_col):
    df['correct'] = df['y_true'] == df['y_pred']
    return df.groupby(race_col)['correct'].mean()

```

```

accuracy_by_group(X_test_copy, 'race_African-American')

[126]
0s
# Add race back from original df using index
X_test_copy['race'] = df.loc[X_test.index, 'race'].values

# disaggregated metrics function
def metrics_by_group(df, group_col):
    df['correct'] = df['y_true'] == df['y_pred']
    return df.groupby(group_col).agg(
        accuracy=('correct', 'mean'),
        count=('correct', 'count')
    )

# Disaggregated accuracy by racial group
# Note: Asian (n=5) and Native American (n=3) have small sample sizes
metrics_by_group(X_test_copy, 'race')

[127]
0s
#race back to test set
X_test_copy['race'] = df.loc[X_test.index, 'race'].values

#fairness metrics function
def fairness_metrics(df, group_col, group1, group2):
    g1 = df[df[group_col] == group1]
    g2 = df[df[group_col] == group2]

    results = pd.DataFrame({
        'metric': ['demographic_parity', 'false_positive_rate', 'false_negative_rate',
'predictive_parity'],
        group1: [
            g1['y_pred'].mean(),
            (g1[(g1['y_true'] == 0) & (g1['y_pred'] == 1)].shape[0] / g1[(g1['y_true']
== 0)].shape[0]),
            (g1[(g1['y_true'] == 1) & (g1['y_pred'] == 0)].shape[0] / g1[(g1['y_true']
== 1)].shape[0]),
            (g1[(g1['y_true'] == 1) & (g1['y_pred'] == 1)].shape[0] / g1[(g1['y_pred']
== 1)].shape[0])
        ],
        group2: [
            g2['y_pred'].mean().round(3),
            (g2[(g2['y_true'] == 0) & (g2['y_pred'] == 1)].shape[0] / g2[(g2['y_true']
== 0)].shape[0]),
            (g2[(g2['y_true'] == 1) & (g2['y_pred'] == 0)].shape[0] / g2[(g2['y_true']
== 1)].shape[0]),
            (g2[(g2['y_true'] == 1) & (g2['y_pred'] == 1)].shape[0] / g2[(g2['y_pred']
== 1)].shape[0])
        ]
    })
    return results

# Phase 2: Fairness Metrics
# Computes demographic parity, FPR, FNR, and predictive parity
# for African- American vs Caucasian defendants
fairness_metrics(X_test_copy, 'race', 'African-American', 'Caucasian')

[128]
6s
!pip install shap -q

```

```

import shap

[129]
9s
X_test_sample = X_test.sample(200, random_state=42)

explainer = shap.TreeExplainer(model)
shap_explanation = explainer(X_test_sample)

# Global feature importance bar chart - shows top features by mean SHAP value
shap.summary_plot(shap_explanation[:, :, 1], X_test_sample, plot_type="bar")

[130]
0s
# Find a Black defendant predicted high risk who did not reoffend (false positive)
X_test_copy['race'] = df.loc[X_test.index, 'race'].values

# Recourse test: find a black defendant wrongly labeled high risk
# y_true0 (did not reoffend) but y_pred=1 (predicted high risk)
false_positives = X_test_copy[
    (X_test_copy['race'] == 'African-American') &
    (X_test_copy['y_true'] == 0) &
    (X_test_copy['y_pred'] == 1)
]

# Pick the first one
individual = false_positives.iloc[0]
print(individual)
age                22
priors_count        3
juv_fel_count       0
juv_misd_count       0
sex_Female           False
sex_Male             True
race_African-American True
race_Asian           False
race_Caucasian       False
race_Hispanic        False
race_Native American False
race_Other           False
y_true              0
y_pred              1
correct              False
race                African-American
Name: 676, dtype: object

[131]
0s
# Test what changes would flip the prediction
individual_features = false_positives.iloc[[0]].drop(
    columns=['y_true', 'y_pred', 'correct', 'race']
)

print("Original prediction:", model.predict(individual_features)[0])

# What if priors_count was 0?
individual_no_priors = individual_features.copy()
individual_no_priors['priors_count'] = 0
print("With 0 priors:", model.predict(individual_no_priors)[0])

# What if they were older?

```

```

individual_older = individual_features.copy()
individual_older['age'] = 40
print("With age 40:", model.predict(individual_older)[0])

# What if race was Caucasian?
individual_white = individual_features.copy()
individual_white['race_African-American'] = False
individual_white['race_Caucasian'] = True
print("If Caucasian:", model.predict(individual_white)[0])
    Original prediction: 1
    With 0 priors: 1
    With age 40: 1
    If Caucasian: 1

[132]
0s
# What if multiple things changed at once?
# Test single-variable changes – none alone flip the prediction
# Only combining zero priors + older age changes the outcome
# Conclusion: no actionable path to recourse exists for this individual
individual_combined = individual_features.copy()
individual_combined['priors_count'] = 0
individual_combined['age'] = 40
individual_combined['race_African-American'] = False
individual_combined['race_Caucasian'] = True
print("0 priors + age 40 + Caucasian:", model.predict(individual_combined)[0])

# Trying just age + priors
individual_age_priors = individual_features.copy()
individual_age_priors['priors_count'] = 0
individual_age_priors['age'] = 35
print("0 priors + age 35:", model.predict(individual_age_priors)[0])
    0 priors + age 40 + Caucasian: 0
    0 priors + age 35: 0

[133]
0s
# Age influence analysis: recidivism rate by age group# younger defendants reoffend at
significantly higher ratesage_influence=
df.groupby("age_cat")["two_year_recid"].mean()*100age_influence

[134]
0s
# What drives Caucasian predictions vs African-American
shap.summary_plot(shap_explanation[:, :, 1], X_test_sample)

[135]
0s
# Additional Analysis: priors_count as racial proxy
# Black defendants average 71% more prior convictions than white defendants
priors_by_race =
df.groupby('race')['priors_count'].mean().sort_values(ascending=False)
priors_by_race

[136]
0s
# Compare COMPAS scores vs actual recidivism rates by race
# shows COMPAS over-scores black defendants relative to actual outcomes
compas_vs_actual = df.groupby('race').agg(

```

```

    avg_compas_score=('decile_score', 'mean'),
    actual_recidivism=('two_year_recid', 'mean')
).round(3)
compas_vs_actual['actual_recidivism'] = (compas_vs_actual['actual_recidivism'] *
100).round(1)
compas_vs_actual

```

Next steps:

```

[137]
0s
X_test_copy['sex'] = df.loc[X_test.index, 'sex'].values

# Intersectional false positive analysis by race and gender
fp_intersectional = X_test_copy[
    (X_test_copy['y_true'] == 0) &
    (X_test_copy['y_pred'] == 1)
].groupby(['race', 'sex']).size().unstack().fillna(0)

fp_intersectional

```

Next steps:

```

[138]
0s
# =====
# Requirements Summary
# =====
# pandas
# numpy
# scikit-learn
# matplotlib
# seaborn
# shap
#
# To reproduce: Run all cells top to bottom
# Data loads automatically from GitHub URL - no local files needed
# random_state=42 used throughout for reproducibility
# =====

[138]

```